

Announcements

- none

Today

- Finish off simulation for number of events over time (emergent Poisson distribution)
- Multiple event types in continuous time
- Gillespie algorithm

Two possible event types

- Find prey (rate s)
- Death (rate m)
- At any moment, either might happen
- Which happens next?

Two possible event types

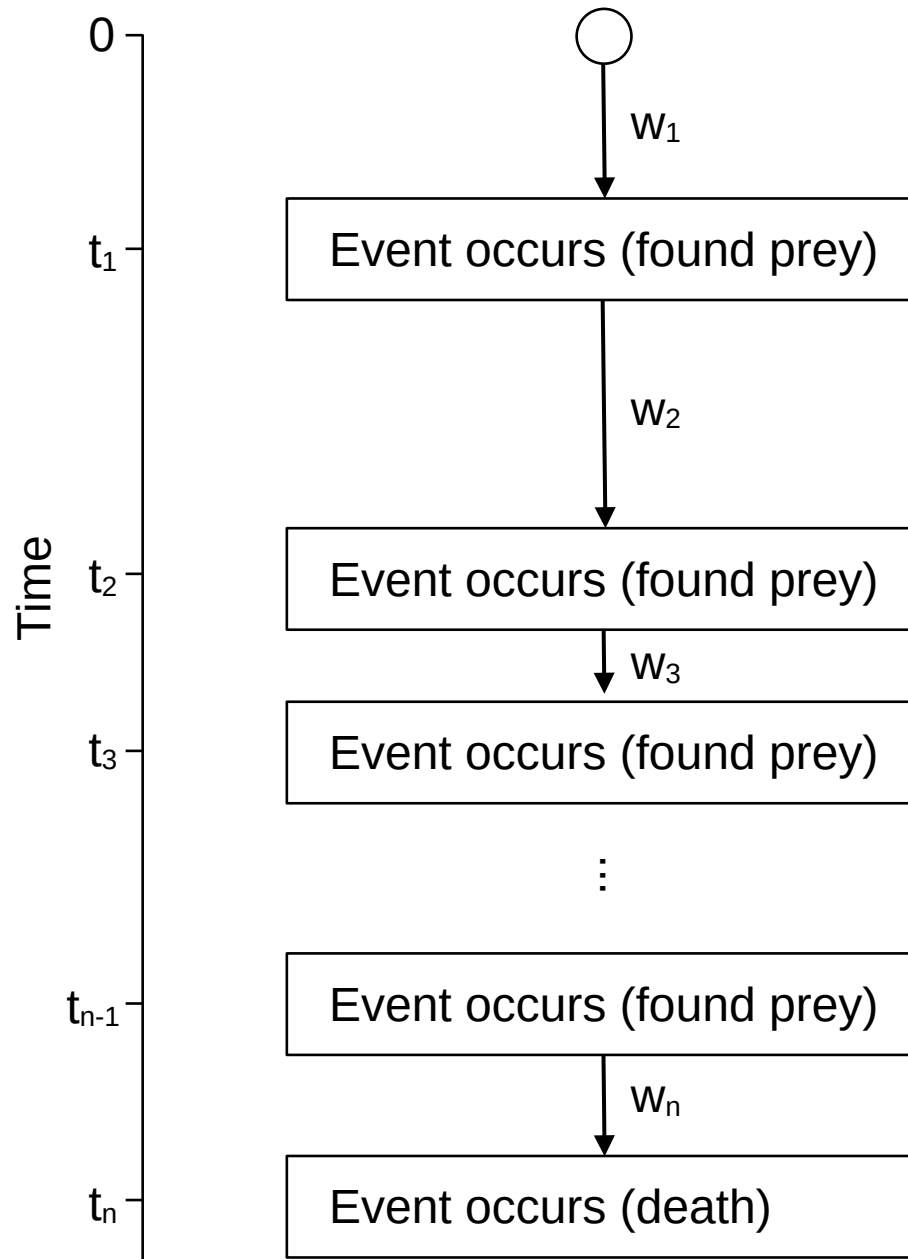
- If searching and death are independent
- $\Pr(\text{either event}) = s\Delta t + m\Delta t$
- $\lim \Delta t \rightarrow 0$:
- Time to next event (waiting time):
- $W \sim \text{Exponential}(s + m)$

Two event types

Event
sequence
diagram

t: event time
w: waiting time

$$w \sim \text{Exp}(s + m)$$



Foraging + death

Algorithm

set s, m

set t = 0, prey = 0, alive = true

while alive

 draw waiting time w, $\text{Exp}(s + m)$

 t = t + w

 if event type is prey found, $p = s / (s + m)$

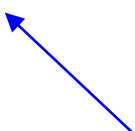
 prey = prey + 1

 else

 alive = false

print t, prey

Bernoulli(p)



Simulate multiple event types

r_i : birth, death, immigration, predation, mutation ...

Algorithm (Gillespie)

intensity = $r_1 + r_2 + \dots + r_n$

repeat

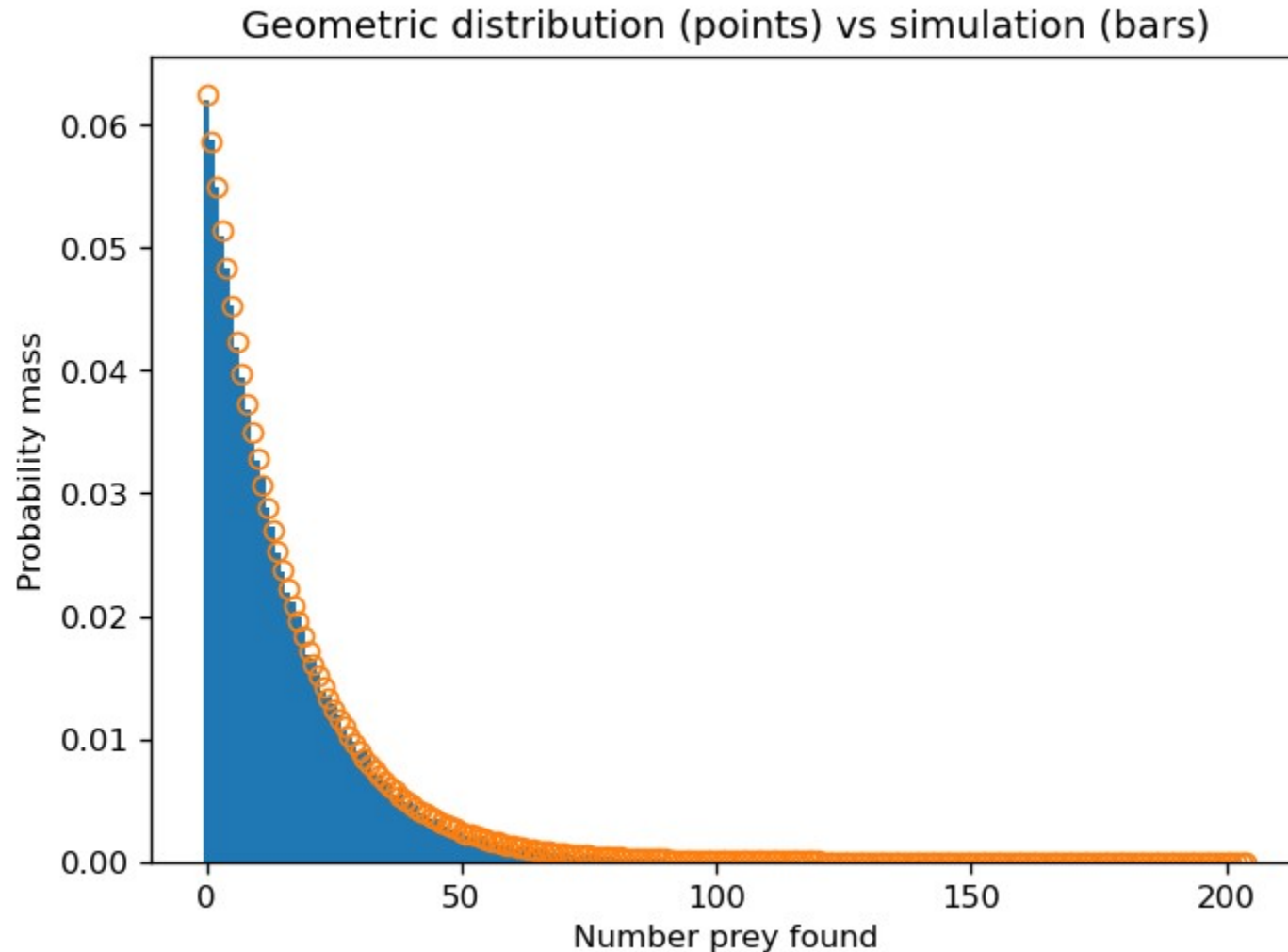
 draw waiting time w from $\text{Exp}(\text{intensity})$

$t = t + w$

 decide which event occurred, $p_i = r_i / \text{intensity}$

 update state

Emergent distribution (geom)



Geometric distribution

- In this case, emergence is very neat
- N = number of prey

$$\Pr(N = n) = (p_{prey})^n (p_{death})$$

$$= \left(\frac{s}{s+m} \right)^n \left(\frac{m}{s+m} \right)$$



Parameters of the 2-event
Poisson process

Geometric distribution

- Emerges as the **sum of successes** before the **first failure**
- Consequence of the **sequence** of events, where prey is success and death is failure

prey prey prey prey prey **death**

In this case, the Poisson process (waiting times) determines the relative probability of success vs failure

Geometric distribution

- X is the number of failures before success
- p = probability of success

$$\Pr(X = x | p) =$$

Discrete x
 $0, 1, 2, \dots, n$

$$f(x, p) = (1 - p)^x p$$

PMF

$$X \sim \text{Geom}(p)$$

Geometric distribution

$$E[X] = \frac{(1-p)}{p}$$

$$\text{Var}[X] = \frac{(1-p)}{p^2}$$